

Data Management
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Master degree on Data Science
Academic Year 2022/2023
17 January 2023

60 minutes

Question 1

We want to build a dashboard that publishes the results on air quality in Milan every day. Imagine that the Lombardy region provides a page, updated every hour with all the data from the stations that monitor air quality (for Milan, assume only two stations are available). It is also possible to access the data history of the previous day via API. Finally, there is a network of volunteers who make available the control units installed outside their homes in Milan (about 120 control units) which can be accessed via API.

Define a possible strategy for the acquisition of air quality data from one or more of the indicated data sources based on the quality, precision, accuracy, and updating characteristics required to obtain the dashboard

Question 2

To build a dance venue review platform it is necessary to build a database where there are the list of venues, user comments, opinions on the comments and the list of registered users with their comments. Define, motivating it, the most appropriate data model taking into account the specifications, scalability and availability characteristics that the application must have

Question 3

Describe the differences in use between a data lake architecture and a data warehouse